

Reducing Regressivity in ML-Based Mass Appraisal: A Covariance-Guided Correction for Assessor Workflows

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Abstract

Property value assessments determine how local property-tax burdens are distributed. Assessor offices increasingly use machine learning-based automated valuation models to estimate values for large and varied property inventories, but strong average prediction does not prevent *regressivity*: lower-value properties can receive systematically higher assessment-to-sale ratios than higher-value properties. Working with data scientists and assessors at the Cook County Assessor’s Office (CCAO), we develop a correction that can be added to its existing LightGBM residential workflow without replacing its data, features, or review process. The method adds a training penalty for a systematic relation between assessment errors and sale prices. We consider a direct squared-covariance penalty and a sample-additive, covariance-guided objective that is compatible with standard boosting software. Applied to a Python translation of the CCAO workflow, the validation-selected sample-additive model substantially improves three standard measures of vertical equity: their departures from neutral values fall by about 54% to 88% relative to ordinary LightGBM. Predictive fit changes little, while mean absolute error rises by about 3.5%. The current comparison measures the full implemented custom-objective path and does not yet isolate the penalty from every implementation difference. The correction targets a global first-order pattern. It does not set the overall assessment level, remove local inequity, or guarantee that every ratio-study measure will fall within its reference range. It should therefore be used with temporal validation, uncertainty analysis, subgroup ratio studies, and comparison with simpler post-hoc recalibration.

1 Introduction

Property taxes are one of the largest sources of local government revenue in the United States. They help finance public schools, roads and streets, and other essential local services [23]. Using 2016 U.S. Census Bureau data, Berry [4] reports that property taxes accounted for roughly 72% of local tax revenue and 47% of local own-source general revenue, and totaled about \$500 billion annually.

Property taxation in the United States is generally based on property values: local assessors estimate property values, which are then used, together with jurisdiction-specific assessment and taxation rules, to determine each taxpayer’s share of the property-tax burden [23, 25]. Consequently, systematic overassessment of some properties and underassessment of others can shift tax liability from one group of property owners to another, creating or exacerbating inequities [4, 23].

The valuation of groups of properties for tax purposes is known as *mass appraisal*. The International Association of Assessing Officers (IAAO) defines mass appraisal as “the process of valuing a group of properties as of a given date using common data, standardized methods, and statistical testing” [25]. Within this process, assessors use statistical models to estimate property values across large property groups. When implemented in computer-based valuation systems, such models are commonly referred to as automated valuation models (AVMs) [19, 5]. Recent mass-appraisal research increasingly examines nonlinear machine-learning (ML) methods, including gradient-boosted trees, alongside traditional regression approaches [36, 38, 43].

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However, when evaluating the performance of a mass-appraisal valuation model, predictive accuracy alone is insufficient. The IAAO recommends the use of *ratio studies*, which compare appraised values with observed sale prices. A central quantity in these studies is the ratio of appraised value to sale price. Because the CCAO model studied here predicts *fair market value* before the statutory conversion to assessed value, we evaluate its predictions on a common market-value scale, using the model-estimated market value divided by the observed sale price; we refer to this quantity as the *assessment ratio* throughout the paper [12]. Ratio studies use such ratios to evaluate assessment level and uniformity, including whether similar properties are treated consistently (*horizontal equity*) and whether assessment ratios vary systematically across property-value levels (*vertical equity*) [21, 25].

A particularly relevant form of vertical inequity is assessment *regressivity*: lower-value properties tend to receive higher assessment ratios than higher-value properties. This pattern can place a disproportionate share of the assessment, and potentially tax, burden on owners of lower-valued properties [32, 4, 3]. Moreover, regressivity may remain hidden in aggregate measures of predictive accuracy because these measures do not capture whether assessment errors vary systematically with property value. A valuation model can therefore achieve strong predictive accuracy while still exhibiting systematic price-related bias and, consequently, inequitable assessment outcomes.

This motivates treating regressivity as an explicit criterion in model training or calibration rather than only as an ex-post diagnostic. In mass appraisal, a useful correction must operate within existing sales-based modeling, validation, and review processes: current prices are observed only for properties that transact, while valuation systems must produce values within fixed assessment schedules and remain compatible with professional and administrative review [24, 22, 25]. Accordingly, the correction should reduce regressivity while preserving predictive performance and requiring minimal changes to existing valuation workflows.

This paper arises from a collaboration between the Cook County Assessor’s Office (CCAO) and academic researchers, motivated by an operational goal: reducing regressivity in the CCAO’s residential valuation workflow. The CCAO currently uses a county-wide LightGBM (Light Gradient-Boosting Machine) model [27] as its primary residential valuation model, following earlier township-level linear-regression and gradient-boosting workflows [12]. The current model is trained on screened residential sales and estimates a *fair market value* using property characteristics, location, environmental variables, and market trends. It operates within a broader assessment pipeline that separates data ingestion, model training, assessment, evaluation, interpretation, and finalization [12].

The model’s fair-market-value estimate is only one input into the final property-tax bill. In Cook County, that value is converted to assessed value according to property classification, equalized using the state equalization factor, reduced by eligible exemptions, and ultimately combined with local tax rates determined from taxing-district levies and taxable property values [14, 13]. We therefore focus on the valuation stage rather than the downstream steps that convert estimated market values into final tax liabilities.

Within the valuation stage, the central problem addressed in this paper is the interaction between predictive performance and regressivity. In the CCAO workflow considered, the gradient boosting model substantially improves predictive accuracy and ratio dispersion relative to the linear-regression baseline, but its vertical-equity diagnostics move in a more regressive direction. This creates a model-design problem: rather than choose between the linear and gradient-boosting baselines, we seek to retain the predictive advantage of gradient boosting while explicitly reducing regressivity. From an operational perspective, the correction should also fit within CCAO’s existing modeling and validation workflow without requiring a fundamental change to CCAO’s existing modeling and validation workflow [12].

We address this challenge by modifying the training objective of the valuation model so that regressivity becomes an explicit model-design consideration. Specifically, we formulate a direct squared-covariance penalty that controls global price-related dependence between valuation errors and property values, and then develop a sample-additive surrogate motivated by the same dependence-control objective that is compatible with LightGBM’s custom-objective interface. For a fixed penalty formulation, the correction changes only the training objective and introduces one additional tuning parameter: the penalty strength. The correction requires no changes to the underlying data, feature set, or surrounding validation and evaluation workflow. The penalty strength can be selected through chronological validation using the predictive and ratio-study measures already used to evaluate mass-appraisal models. The resulting approach is therefore designed to reduce regressivity while remaining compatible with the existing machine-learning valuation pipeline. The framework is intentionally targeted: it does not, by itself, determine overall assessment level or guarantee

local or subgroup equity.

Applied to CCAO’s residential property-valuation workflow, the proposed methodology substantially reduces measured regressivity while preserving nearly all of the predictive performance of the unpenalized LightGBM. We use sales data from 2016–2023 for model development, evaluate the selected models on a held-out 2024 test set, and use 2025 as a secondary forward evaluation aligned with the assessment-year workflow. Relative to unpenalized LightGBM, the validation-selected model reduces the deviations of three complementary vertical-equity diagnostics from their neutral values by approximately 54% to 88%. At the same time, R^2 decreases only from 0.885 to 0.881, while mean absolute error increases by approximately 3.5%. Thus, the proposed methodology produces a substantial reduction in measured regressivity at a small cost in predictive accuracy. Exploratory experiments across six counties further suggest that the correction can improve value-related assessment patterns in different markets, although the appropriate penalty strength and attainable improvement vary across jurisdictions.

The method was developed in collaboration with CCAO data scientists and has been incorporated into the office’s residential model-development codebase. [12] The CCAO has also used and tested the correction during model development. This implementation provides direct evidence that the proposed approach can be integrated into an existing public-sector mass-appraisal workflow without requiring a fundamental redesign of the underlying valuation pipeline.

The paper makes three contributions. First, motivated by an important operational problem recognized by CCAO, ~~we translate property tax regressivity into a tractable model-training objective that explicitly targets the relationship between valuation errors and property values.~~we connect assessor-defined assessment regressivity with dependence-based constrained learning by translating the value-related error pattern into a tractable model-training objective. Second, ~~we develop an easy-to-incorporate implementation that operates through the standard LightGBM interface and fits within an assessor’s existing temporal-validation and ratio-study workflow.~~we develop a direct squared-covariance penalty and a sample-additive surrogate that can be implemented through the standard LightGBM interface and evaluated within an assessor’s existing temporal-validation and ratio-study workflow. Third, we demonstrate that the correction can substantially reduce regressivity while largely preserving predictive accuracy, and ~~we document its progression from a jointly developed research method to a component of CCAO’s core residential valuation codebase.~~we document its incorporation into CCAO’s residential model-development codebase.

The paper makes three main contributions. First, we connect assessor-defined assessment regressivity with dependence-based constrained learning by formulating assessment regressivity as systematic dependence between valuation errors and property value. This provides a direct way to translate a standard mass-appraisal equity concern into an explicit model-training objective. Second, we develop a direct squared-covariance penalty and a sample-additive surrogate motivated by the same dependence-control objective, and show how these formulations can be incorporated into regression models used in mass appraisal, including linear regression and gradient boosting. The penalty strength is selected through chronological validation using predictive and ratio-study criteria. Third, using CCAO’s residential valuation workflow, we show that the proposed correction can substantially reduce measured regressivity with limited loss in predictive accuracy. The method has also been implemented and tested within CCAO’s model-development pipeline, providing evidence of its operational compatibility with a large public-sector mass-appraisal system.

The remainder of this section reviews the related literature on machine learning in mass appraisal, vertical equity, and dependence-based fair regression. The rest of the paper is organized as follows. Section 2 describes the mass-appraisal setting, the CCAO valuation workflow, and the evaluation framework. Section 3 develops the proposed covariance-based regressivity correction. The remaining sections present the empirical evaluation, examine the robustness and practical implementation of the approach, and discuss its limitations and extensions.

1.1 Related Work

Machine Learning in Mass Appraisal Recent reviews document the growing research on automated valuation models and machine learning (ML) in mass appraisal and real estate valuation. Multiple regression remains common, alongside tree ensembles such as random forests and gradient boosting, as well as neural-network-based methods [41, 36, 43, 2]. Most studies compare these methods primarily through predictive criteria, such as out-of-sample accuracy and performance relative to alternative algorithms. Vertical equity,

Update all numerical results after the final experimental run. Add confidence intervals? Is mean enough?

Replace or supplement this citation with the specific CCAO commit/PR containing the correction.

I think we need to discuss this further. I see as a contribution: (1) a bridge between assessors literature and fairness in ML, (2) and implementation in a public workflow proved.

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although routinely evaluated in assessor practice, is much less often incorporated directly into model training or model selection [41, 36, 43]. When predictive performance is the primary criterion, tree ensembles are among the most frequently studied and often strong-performing nonlinear methods in this literature [36, 38, 43]. A practical training-time equity intervention should therefore be compatible with these commonly studied and used model classes rather than require a specialized valuation architecture.

Operational adoption by assessment offices nevertheless remains partial and uneven. Bidanset and Rakow [5] identify communication challenges, limited technical capacity, and budget constraints among perceived barriers to wider adoption. Moreover, predictive modeling is only one component of the assessment workflow; the use of ML does not replace sales verification, out-of-time evaluation, ratio studies, model interpretation, or administrative review [24, 19]. The IAAO Artificial Intelligence Task Force [19] emphasizes explainability, defensibility, administrative feasibility, and integration into existing workflows as central conditions for operational use of advanced ML models. These considerations favor equity-aware methods that can be incorporated into existing model-training, validation, and review pipelines with minimal disruption.

Vertical equity and regressivity. Equity has long been a central objective in mass appraisal practice and standards. The assessor literature distinguishes *horizontal equity*, which concerns the uniform treatment of similar properties, from *vertical equity*, which concerns systematic differences in assessment outcomes across property-value levels [21, 15, 33]. A common form of vertical inequity is *regressivity*: assessment-to-sale ratios tend to decline with property value, so lower-valued properties tend to be relatively overassessed and higher-valued properties tend to be relatively underassessed. Ratio studies evaluate assessment uniformity using complementary diagnostics. For example, the Coefficient of Dispersion (COD) summarizes dispersion in assessment ratios, whereas the Price-Related Differential (PRD) and Price-Related Bias (PRB) are commonly used to diagnose price-related vertical inequity [21, 20].

No single diagnostic fully characterizes vertical equity. Methodological studies show that vertical-equity measures can differ materially in their statistical behavior, and that some regression-based diagnostics can be biased when assessed values are themselves generated by regression-based valuation models [20, 33]. Accordingly, the assessor literature recommends examining multiple complementary diagnostics rather than relying on a single global measure. Empirical studies further document substantial regressivity and consequential distributional disparities across jurisdictions [4, 3]. Recent large-scale evidence also shows that predictive accuracy and assessment equity need not be intrinsically opposed: improvements in the information available to valuation models can, in many settings, improve both simultaneously [39].

A smaller literature moves beyond aggregate diagnosis toward model specification, localized validation, and explicit correction intended to improve assessment equity. Proposed approaches include spatial and geographically weighted methods, stratified procedures, ratio-study-based model validation, and fairness-aware valuation models [35, 6, 7, 17, 18, 8]. Most closely related to our work, Candogan, Han, and Lu [8] develop a *K-segment* property-valuation model that explicitly targets regressivity and studies the associated accuracy–fairness relationship. Our approach addresses the same assessor-side objective through a different mechanism: rather than introducing a segmented valuation architecture, we incorporate a dependence-based penalty directly into the training objective of existing regression learners. The method developed here therefore contributes to this strand by directly targeting a global price-related dependence associated with regressivity during training, while retaining standard ratio studies and subgroup review for out-of-sample evaluation.

Fair regression and dependence-based constrained learning. Although much of the fairness-in-ML literature initially focused on classification, *fair regression* studies how fairness requirements can be incorporated when predictions are continuous, as in property valuation. Existing approaches include constrained or regularized empirical risk minimization and post-processing methods. Different works consider fairness notions such as statistical parity, bounded group loss, mean parity, or distributional parity. Methodologically, these requirements can be imposed through constrained or regularized empirical risk minimization or through post-processing and recalibration procedures [1, 16, 10, 9, 42]. The broader methodological principle is that predictive risk can be optimized subject to, or jointly with, an application-specific fairness criterion. Varying the strength of this requirement produces a family of solutions that makes the relationship between predictive performance and the targeted fairness criterion explicit, without requiring that improvements in one necessarily worsen the other.

Do we need to sparsify or make more granular comments here? I feel I aggregated all the literature in one chunk

A particularly relevant strand expresses fairness requirements as restrictions on statistical dependence between predictions and a sensitive variable. Prior work has quantified or controlled such dependence using correlation and coefficients of determination [28], kernel-based measures such as the Hilbert–Schmidt Independence Criterion (HSIC) [34, 30], and maximal-correlation criteria [31, 29]. Related regularization approaches also control the contribution of sensitive attributes to regression predictions [37]. These approaches are particularly relevant when the variable of concern is continuous and can often be incorporated directly into regression objectives. Our application, however, differs from conventional group-fairness settings. Sale price, our observed proxy for market value, is the prediction target rather than a protected attribute, and our objective is not demographic parity. We instead adapt the general dependence-control principle to a different structural relationship: dependence between valuation errors and property value. The assessor literature therefore supplies the domain-specific target—price-related regressivity—while the fair-regression literature supplies the general dependence-control mechanism. The proposed method connects the two by translating value-related dependence in valuation errors into a trainable regularization objective.

2 Preliminaries: The CCAO Baseline and Empirical Motivation

This section fixes the CCAO application setting, prediction notation, and evaluation framework used throughout the paper. We first describe the residential valuation workflow and baseline model classes. We then distinguish predictive performance, assessor-facing measures of assessment level and ratio dispersion, complementary vertical-equity diagnostics, and supplemental checks for nonlinear or localized value-related patterns. Finally, we present the controlled baseline comparison that motivates the proposed correction.

2.1 The CCAO Residential Valuation Workflow

The CCAO values a large and heterogeneous residential inventory on a triennial reassessment cycle [11]. Its residential modeling evolved from township-level linear-regression models, with linear-regression and gradient-boosting alternatives used in 2019–2020, to a county-wide LightGBM model beginning in 2021 [12]. LightGBM is an implementation of gradient-boosted decision trees [27].

The CCAO’s current published residential automated valuation model (AVM) uses verified sales and a broad set of property, spatial, neighborhood, market, and contextual predictors to estimate residential market values [12]. The prediction model is embedded in a broader operational workflow comprising data preparation, model training, valuation, evaluation, interpretation, and final review. Candidate valuations are evaluated using both predictive-performance measures and assessor-specific ratio-study statistics across multiple geographic areas and property classes.

For the empirical analysis, we compare a linear-regression benchmark motivated by CCAO’s earlier models with unpenalized LightGBM, representing its current model class, while holding fixed the prediction target, underlying feature information, development sample, temporal design, and evaluation procedures. The comparison is therefore a controlled comparison of model classes rather than a reconstruction of official historical assessments. The exact sample construction, transaction filters, and temporal validation procedures are described in Section 4.1.

2.2 Prediction Setting and Baseline Models

Let $\mathcal{I}$ denote the set of residential sales retained after the CCAO sales-validation and transaction-eligibility procedures described in Section 4.1. For each $i \in \mathcal{I}$, we observe

$$(x_i, P_i, t_i),$$

where x_i contains property and parcel characteristics; tract-level socioeconomic and demographic variables from the U.S. Census Bureau’s American Community Survey (ACS); and geographic, neighborhood, environmental, accessibility, market, and temporal features.

The variable $P_i > 0$ is the verified sale price and is treated as the observed transaction-based proxy for the property’s market value, while t_i is the sale date used to construct chronological samples and time-related predictors [12, 26]. Market value itself is latent: a verified sale therefore provides an observed market-value

decide if the t represents any predictor, since they come basically from the same original feature

indicator rather than a directly observed market value. The prediction problem is to estimate market value from the information available in x_i .

Since residential sale prices are positive and highly right-skewed, the baseline models are trained on the log-price scale. Define

$$y_i = \log P_i, \quad \hat{y}_i = f(x_i), \quad e_i(f) = \hat{y}_i - y_i, \quad (1)$$

where $f \in \mathcal{H}$ is a candidate prediction function. The corresponding prediction on the original price scale is

$$\hat{P}_i = \exp(f(x_i)). \quad (2)$$

The log-price scale is the estimation scale, whereas $\hat{P}_i$ is the directly retransformed valuation used for price-scale predictive evaluation and ratio-study diagnostics. Throughout the analysis, $\hat{P}_i$ is defined operationally by (2). Note that the direct exponentiation should not be interpreted as a claim that $\hat{P}_i$ is a bias-corrected estimate of the expected value of property i given x_i . We define the corresponding *valuation-to-sale ratio* as

$$r_i(f) = \frac{\hat{P}_i}{P_i} = \exp(e_i(f)). \quad (3)$$

For brevity and consistency with ratio-study terminology, we subsequently refer to $r_i(f)$ as the *assessment ratio*. Its numerator is the model-estimated market value, not Cook County's statutory assessed value after downstream adjustments. When a fitted model is fixed, we suppress the dependence on f and write simply r_i and e_i .

The log-scale residual and the assessment ratio are therefore linked exactly by

$$e_i(f) = \log \hat{P}_i - \log P_i = \log r_i(f). \quad (4)$$

Thus, $e_i(f) > 0$ (equivalently, $r_i(f) > 1$) corresponds to a prediction above the observed sale price, whereas $e_i(f) < 0$ (equivalently, $r_i(f) < 1$) corresponds to a prediction below it. Estimation on the log-price scale therefore converts multiplicative price errors into additive residuals, and the training residual is exactly the log assessment ratio.

For the predictive models, we consider two baseline model classes: linear regression and gradient-boosted decision trees. As described above, these choices provide a historically motivated linear benchmark and a representation of CCAO's current model class. Between models, we hold fixed the prediction target, development observations, underlying feature information, temporal design, and evaluation protocol; the models differ in functional form and model-specific estimation procedure.

For a training sample of size n , both baselines are fitted to reduce mean squared log-price error, i.e.,

$$L(f) = \frac{1}{n} \sum_{i=1}^n e_i(f)^2, \quad (5)$$

while differing in functional form, estimation procedure, and model-specific complexity control. This loss is the predictive component of the training objective modified in Section 3.

For linear regression,

$$f_{\text{lin}}(x) = \beta_0 + x^\top \beta. \quad (6)$$

The hypothesis class is therefore restricted to functions that are linear in the supplied predictors. In the ordinary linear-regression baseline, no additional regressivity term is included in the training objective. This specification provides the controlled linear benchmark motivated by CCAO's earlier linear-regression models. Its linear structure imposes a comparatively rigid relationship between the predictors and log sale price.

On the other hand, LightGBM, the modeling foundation of CCAO's current published residential AVM, constructs an additive ensemble of regression trees [12, 27]. A gradient-boosted tree predictor can be written as

$$f_{\text{gbm}}^{(M)}(x) = f^{(0)}(x) + \sum_{m=1}^M \nu b_m(x), \quad (7)$$

where $f^{(0)}$ is an initial prediction, b_m is the regression tree added at boosting iteration m , M is the number of boosting iterations, and $\nu > 0$ is the learning rate. Unlike the linear model, the tree ensemble can represent nonlinearities, threshold effects, and interactions without requiring them to be specified in advance.

re-think about this name. The IAAO uses ratios-to-something, and assessments can be confusing given the adjustments applied to the prediction, to get the assessment value.

Confirm if I defined AVM or if I should use another term instead

LightGBM fits the ensemble in (7) sequentially. The gradient and Hessian calculations required for the proposed custom objectives are introduced with the penalties in Section 3.

The two baselines therefore solve the same prediction problem under a common empirical design but use substantially different model classes. Both baseline objectives minimize squared log-price error and contain no term that directly controls systematic value-related patterns in residuals or assessment ratios. We next distinguish predictive performance from the assessor-facing criteria used to evaluate assessment level, ratio dispersion, and regressivity.

2.3 Predictive and Assessor-Facing Evaluation Criteria

The two scales play distinct roles in evaluation. Predictive performance is assessed both on the log-price scale used for estimation and on the original price scale on which valuations are reported. Assessor-facing ratio-study diagnostics are computed on the original price scale. The resulting measures answer different questions and are therefore interpreted jointly rather than collapsed into a single performance criterion.

Predictive performance. For an evaluation sample $S \subseteq \mathcal{I}$, let $n_S = |S|$ and $\bar{P}_S = \frac{1}{n_S} \sum_{i \in S} P_i$. We use four complementary headline accuracy-related measures: price-scale mean absolute error (MAE), price-scale mean absolute percentage error (MAPE), price-scale R^2 , and log-scale root mean squared error (RMSE):

$$\text{MAE}_P(S) = \frac{1}{n_S} \sum_{i \in S} |\hat{P}_i - P_i|, \quad (8)$$

$$\text{MAPE}_P(S) = 100 \times \frac{1}{n_S} \sum_{i \in S} \left| \frac{\hat{P}_i - P_i}{P_i} \right| = 100 \times \frac{1}{n_S} \sum_{i \in S} |r_i - 1|, \quad (9)$$

$$R_P^2(S) = 1 - \frac{\sum_{i \in S} (\hat{P}_i - P_i)^2}{\sum_{i \in S} (P_i - \bar{P}_S)^2}, \quad (10)$$

$$\text{RMSE}_{\log P}(S) = \left(\frac{1}{n_S} \sum_{i \in S} e_i(f)^2 \right)^{1/2}. \quad (11)$$

The price-scale MAE gives the average absolute prediction error in dollars, while MAPE gives the average absolute error relative to the observed sale price. In particular, MAPE measures deviation from the economically neutral ratio $r_i = 1$ and is therefore distinct from COD below, which measures dispersion around the sample median ratio. The statistic R_P^2 summarizes predictive fit on the original price scale. Finally, $\text{RMSE}_{\log P}$ directly evaluates the scale and squared-error loss used for estimation and, by (4), is equivalently the root mean squared log-ratio error.

Assessment level. Recall from (3) that $r_i = \hat{P}_i / P_i$. A ratio above one indicates that the predicted value exceeds the observed sale price, whereas a ratio below one indicates that it falls below the sale price. Define the median, arithmetic mean, and price-weighted mean ratios as

$$m_S = \text{median}_{i \in S} \{r_i\}, \quad \bar{r}_S = \frac{1}{n_S} \sum_{i \in S} r_i, \quad \bar{r}_{W,S} = \frac{\sum_{i \in S} \hat{P}_i}{\sum_{i \in S} P_i} = \frac{\sum_{i \in S} r_i P_i}{\sum_{i \in S} P_i}. \quad (12)$$

These quantities summarize aggregate assessment level. Because both numerator and denominator of r_i are expressed on the market-value scale in this analysis, the neutral reference value for the three statistics is a value of one. The median is relatively insensitive to extreme ratios, the arithmetic mean assigns equal weight to each sale, and the weighted mean places greater weight on higher-value properties. Agreement among these measures indicates broad aggregate calibration but does not establish either horizontal or vertical equity.

Reconcile the code/output “OOS R2” with the definitions in this subsection. Do we use train mean as baseline or the test one, as usually done?

Ratio dispersion (Horizontal uniformity). Ratio dispersion is an important assessor-facing dimension of horizontal uniformity. We summarize it using the coefficient of dispersion (COD) and coefficient of variation (COV). The COD is

$$\text{COD}(S) = 100 \times \frac{1}{n_S} \sum_{i \in S} \frac{|r_i - m_S|}{m_S}, \quad (13)$$

and the COV is

$$\text{COV}(S) = 100 \times \frac{s_{r,S}}{\bar{r}_S}, \quad s_{r,S} = \left(\frac{1}{n_S - 1} \sum_{i \in S} (r_i - \bar{r}_S)^2 \right)^{1/2}. \quad (14)$$

The COD measures the average absolute deviation from the median ratio, whereas the COV measures standard deviation relative to the mean. Lower values indicate less aggregate ratio dispersion, but the reference ranges depend on property class and prevailing market conditions, and unusually low dispersion can itself warrant diagnostic review [21, 26]. To avoid ambiguity with the covariance penalty introduced later, COV always denotes the coefficient of variation, whereas statistical covariance is written as $\text{Cov}(\cdot, \cdot)$.

Regressivity (Vertical equity). Vertical equity concerns whether assessment ratios vary systematically with property value. A *regressive* pattern arises when lower-value properties tend to have higher assessment ratios than higher-value properties. A *progressive* pattern arises in the opposite direction.

No single statistic fully characterizes vertical equity: different diagnostics summarize different aspects of the relationship between valuations, ratios, and property value and can therefore disagree [20, 26]. We consequently use a complementary suite of ratio-based, regression-based, distributional, and grouped diagnostics rather than treating any single statistic as definitive. The May 2026 IAAO exposure draft places primary emphasis on the grouped VEI and identifies PRB, MKI, and PRD as complementary measures. Given that the document is still in draft form, we use its proposed procedures and ranges as descriptive guidance rather than as final adopted compliance standards.

The *price-related differential* (PRD) is

$$\text{PRD}(S) = \frac{\bar{r}_S}{\bar{r}_{W,S}}. \quad (15)$$

Note that, since the weighted mean ratio assigns greater weight to higher-value properties, a PRD above one tends to arise when higher-value properties have lower assessment ratios than lower-value properties. The 2013 IAAO standard provides a reference range of [0.98, 1.03] for the PRD: values above this range tend to indicate regressivity, whereas values below it tend to indicate progressivity [21].

PRD is a simple and widely used ratio-based diagnostic, but it can be particularly sensitive to influential high-value observations and extreme ratios. Therefore, we interpret it jointly with the other vertical-equity measures rather than as a standalone test [26].

The following regression- and group-based vertical-equity diagnostics require a proxy for the property's value level. Ranking properties only by sale price can mechanically bias ratio-based vertical-equity tests toward finding regressivity because sale price also appears in the denominator of r_i , while using valuation alone can induce bias in the opposite direction. Following IAAO guidance, we therefore use the equal-weight market-value proxy

$$V_i^{\text{proxy}} = \frac{1}{2} \left(P_i + \frac{\hat{P}_i}{m_S} \right), \quad (16)$$

which combines the verified sale price with the median-normalized predicted value to reduce this source of proxy-induced bias [21, 26].

We first report the *price-related bias* (PRB). Let $\tilde{V}_S^{\text{proxy}}$ denote the median of V_i^{proxy} over S . The PRB is the slope in the simple linear regression

$$\frac{r_i - m_S}{m_S} = a_S + \text{PRB}(S) \cdot \log_2 \left(\frac{V_i^{\text{proxy}}}{\tilde{V}_S^{\text{proxy}}} \right) + \varepsilon_i. \quad (17)$$

Negative PRB values indicate that assessment ratios tend to decline as property value increases and are

I think we can improve the notation here for the slope PRB

therefore associated with regressivity. Values between -0.05 and 0.05 are conventionally preferred [21].

As a complementary distributional diagnostic, we report the *Modified Kakwani Index* (MKI). Let $\pi(1), \dots, \pi(n_S)$ index the observations in S in nondecreasing order of sale price, and let

$$q_j = \frac{j - \frac{1}{2}}{n_S}, \quad \bar{P}_S = \frac{1}{n_S} \sum_{i \in S} \hat{P}_i.$$

Define the *concentration index* of predicted values when observations are ordered by sale price and the corresponding *Gini coefficient* of sale prices as

$$C_{\hat{P}|P}(S) = \frac{2}{n_S \bar{P}_S} \sum_{j=1}^{n_S} \hat{P}_{\pi(j)} q_j - 1, \quad G_P(S) = \frac{2}{n_S \bar{P}_S} \sum_{j=1}^{n_S} P_{\pi(j)} q_j - 1, \quad (18)$$

respectively. Then, the MKI is

$$\text{MKI}(S) = \frac{C_{\hat{P}|P}(S)}{G_P(S)}. \quad (19)$$

When there is sufficient price variation, an MKI value of one indicates vertical neutrality, whereas values below one indicate a regressive direction, and values above one indicate a progressive direction [26]. Because MKI orders observations using sale price, it can itself be affected by errors in the market-value proxy, which is another reason to interpret it as one component of the broader diagnostic suite rather than as a definitive test.

Finally, we report the *Vertical Equity Indicator* (VEI), the quantile-grouped vertical-equity measure emphasized in the May 2026 IAAO exposure draft [26]. The draft varies the number of percentile groups with sample size: for the large evaluation samples used in the main countywide analysis ($n_S \geq 501$), observations are divided into deciles according to V_i^{proxy} . Let S_{q_1} and $S_{q_{10}}$ denote the lowest and highest deciles, and let $m_{S_{q_1}}$ and $m_{S_{q_{10}}}$ denote their median assessment ratios. The VEI is

$$\text{VEI}(S) = 100 \times \frac{m_{S_{q_{10}}} - m_{S_{q_1}}}{m_S}. \quad (20)$$

A negative VEI indicates that higher-value properties have lower median assessment ratios than lower-value properties. The exposure draft proposes a reference band of $[-10\%, 10\%]$, together with confidence-interval-based testing for point estimates outside that band.

Again, since the May 2026 document remains an exposure draft, we use VEI and the proposed band as descriptive guidance rather than as a final adopted compliance standard. Moreover, note that the first-versus-last quantile comparison in (20) does not necessarily reveal non-monotone patterns within the median ratios distribution. Hence, we also report the median assessment ratio and its 90% confidence interval for every VEI percentile group, together with the overall median ratio.

Table 1 summarizes the principal predictive and assessor-facing criteria used throughout the paper. Predictive metrics quantify different aspects of valuation error, assessment-level statistics summarize aggregate calibration, COD and COV summarize ratio dispersion, and VEI, PRB, MKI, and PRD provide complementary grouped, regression-based, distributional, and ratio-based views of vertical equity. Strong predictive performance or low aggregate dispersion does not by itself imply neutral vertical-equity behavior.

Supplemental diagnostics. The assessor-facing statistics are complemented by three checks designed around failure modes of the proposed first-order correction. First, we examine the full VEI percentile-group profile described above rather than only its first-versus-last summary. Second, we report the *distance correlation*.

$$\text{dCor}(e, y) = \text{dCor}(\log r, \log P), \quad (21)$$

as an omnibus diagnostic of residual value dependence [40]. Under standard moment conditions, population distance correlation equals zero if and only if the variables are distributionally independent ($r \perp P$). Note that independence is stronger than vertical neutrality as conventionally evaluated in ratio studies. For example, distance correlation can respond to price-related changes in residual dispersion even when group

Verify that the implementation of MKI uses the price-ordered concentration index in (18), and not the simplified ratio $\text{Gini}(\hat{P}) / \text{Gini}(P)$, which is not the MKI definition used here.

Note: I think we need this or an equivalent metric to also account for the non-linear distributional independence. We already talked about this in the past, but I am making it explicit here. IAAO metrics are not enough IMO. GOing more granular to locaton or other sensitive attributes buys a lot of credibility too. We can sill move it to Appendix if needed...

Table 1: Predictive and assessor-facing evaluation measures.

Measure	Ideal value	Reference range	Ideal direction	Interpretation
<i>Predictive performance</i>				
R_P^2	1	–	Higer $\uparrow$	Original price-scale goodness of fit.
MAE_P	0	–	Lower $\downarrow$	Mean absolute prediction error in dollars.
$MAPE_P$	0	–	Lower $\downarrow$	Mean absolute prediction error relative to sale price.
$RMSE_{\log P}$	0	–	Lower $\downarrow$	RMSE on the log-price estimation scale.
<i>Assessment level</i>				
Median ratio m_S	1	–	$\rightarrow 1$	Robust aggregate assessment level.
Mean ratio $\bar{r}_S$	1	–	$\rightarrow 1$	Unweighted aggregate assessment level.
Weighted mean $\bar{r}_{W,S}$	1	–	$\rightarrow 1$	Value-weighted assessment level.
<i>Horizontal equity</i>				
COD	–	$[5.0, 15.0]^\dagger$	$\rightarrow$ range	Absolute dispersion relative to the median ratio.
COV	–	–	Lower $\downarrow$	Relative dispersion relative to the mean ratio.
<i>Vertical equity</i>				
PRD	1	$[0.98, 1.03]$	$\rightarrow 1$	Mean-to-weighted-mean ratio comparison.
PRB	0	$[-0.05, 0.05]$	$\rightarrow 0$	Regression-based value-related ratio trend.
MKI	1	–	$\rightarrow 1$	Distributional concentration relative to sale prices.
VEI	0	$[-10, 10]^\dagger$	$\rightarrow 0$	High-versus-low value-group median-ratio difference.

Notes. “–” denotes that no universal ideal or reference range is imposed. Arrows indicate the preferred direction of improvement. PRD and PRB reference ranges follow IAAO guidance [21]. † Reference ranges are from the May 2026 IAAO exposure draft: applicable COD ranges depend on type of property and market profile [26].

medians are flat. Hence, we use dCor only to test whether a (linear) covariance-based correction induces nonlinear dependencies, not as an assessor compliance metric. Third, where sample size and value variation are sufficient, we recompute the principal vertical-equity diagnostics previously defined within prespecified geographic and property strata rather than relying exclusively on countywide averages [26].

Three benchmark principles govern the empirical use of these measures. First, point estimates are treated as descriptive model diagnostics rather than as formal determinations of compliance with IAAO standards: uncertainty is reported where it is material to substantive comparisons, including the confidence intervals accompanying the VEI percentile-group profile [21, 26]. Second, countywide statistics are complemented by prespecified geographic and property-stratum analyses when the available sample and value variation support meaningful comparison. Third, during model development the evaluation statistics are computed on chronological validation folds before the aggregation used for model selection. The full evaluation suite is not itself imposed as a collection of constraints over the model, but as a general diagnostic of its estimates.

2.4 Baseline Comparison and Empirical Motivation

~~In this subsection we compare the historically motivated linear regression benchmark with the current model class used by the CCAO: unpenalized LightGBM.~~

We compare the historically motivated linear-regression benchmark with unpenalized LightGBM, representing CCAO’s current model class.

~~To facilitate a controlled comparison of model classes, we reestimate both models using the same outcome definition, development sample, underlying feature information, and surrounding data-processing and evaluation procedures. The comparison is therefore not a direct historical comparison of official assessments produced in different years. Instead, it evaluates the two model classes under a common empirical design.~~

Both models are estimated and evaluated under the same prediction target, underlying feature information, sample-construction rules, temporal-validation framework, and evaluation procedure. The comparison is therefore a controlled comparison of model classes rather than a reconstruction of official assessments produced under different historical workflows. Section 4.1 provides the complete empirical design.

~~The final empirical design uses sales from 2016–2023 for model development, including all model and~~

hyperparameter selection through chronological validation. Candidate configurations are evaluated on chronological development folds before any across-fold aggregation. The held-out 2024 sample provides the primary out-of-sample baseline comparison and is not used for model or hyperparameter selection. The 2025 assessment-year sample is reserved for the separate forward evaluation reported later and is likewise excluded from selection. Section 4.1 gives the complete temporal design and selection protocol. The unpenalized LightGBM benchmark is tuned entirely within this development framework before the regressivity penalties are evaluated.

Old placeholder table removed: a horizontally-resized two-row table reported an obsolete test-set snapshot and outdated results.

Table 2: Baseline comparison under a common empirical design.

Measure	Preferred direction	Linear regression	Unpenalized LightGBM
Prediction			
R_P^2	$\uparrow$	<i>TBD</i>	<i>TBD</i>
MAE_P	$\downarrow$	<i>TBD</i>	<i>TBD</i>
$MAPE_P$	$\downarrow$	<i>TBD</i>	<i>TBD</i>
$RMSE_{\log P}$	$\downarrow$	<i>TBD</i>	<i>TBD</i>
Assessment level			
Median ratio m_S	$\rightarrow 1$	<i>TBD</i>	<i>TBD</i>
Mean ratio $\bar{r}_S$	$\rightarrow 1$	<i>TBD</i>	<i>TBD</i>
Weighted mean $\bar{r}_{W,S}$	$\rightarrow 1$	<i>TBD</i>	<i>TBD</i>
Ratio dispersion			
<i>Horizontal uniformity</i>			
COD	$\rightarrow$ range	<i>TBD</i>	<i>TBD</i>
COV	$\downarrow$	<i>TBD</i>	<i>TBD</i>
Vertical equity			
<i>Regressivity</i>			
PRD	$\rightarrow 1$	<i>TBD</i>	<i>TBD</i>
PRB	$\rightarrow 0$	<i>TBD</i>	<i>TBD</i>
MKI	$\rightarrow 1$	<i>TBD</i>	<i>TBD</i>
VEI	$\rightarrow 0$	<i>TBD</i>	<i>TBD</i>

Notes. Preferred values, directions, and applicable reference ranges are defined in Table 1. Both models are evaluated on the same held-out sample under the common empirical design in Section 4.1.



(a) Linear regression.



(b) Unpenalized LightGBM.

Figure 1: Assessment-ratio profiles across ordered property-value groups for the two baseline models. Points show group median ratios and intervals show the corresponding 90% confidence intervals. Within each panel, groups are ordered from lower to higher values using that model’s market-value proxy in Eq. (16); the horizontal reference is the model’s overall median ratio. Identical axes are used across panels.

Once the final temporal design is chosen, most one sentence here identifying the held-out sample used for the baseline comparison. Keep the complete development, validation, and forward-evaluation chronology in Section 4.1.

Populate Table 2 after the final empirical design is locked. All entries must be computed from the same held-out predictions and the same final metric implementation.

Generate the two panels in Figure 1 from the same held-out predictions used in Table 2. Within each panel, construct the value groups using that model’s V_i^{pFoxy} from Eq. (16). Report the median ratio and its 90% confidence interval for each group. Use identical axes across panels and show the model’s overall median ratio as the horizontal reference.

Table 2 and Figure 1 apply the evaluation framework in Section 2.3 to the two baseline model classes. Assessment level and vertical equity are interpreted separately: the former concerns aggregate calibration of the ratios, whereas the latter concerns how those ratios vary across property-value levels.

~~The baseline comparison should not be interpreted as evidence that machine-learning models are intrinsically more regressive than linear models.~~

~~Instead, the purpose of the comparison is to identify the application-specific tension motivating the proposed method: improvements in predictive accuracy and aggregate ratio dispersion need not eliminate systematic price-related regressivity. The assessor-facing vertical-equity suite and supplemental robustness diagnostics defined above remain the out-of-sample criteria for evaluating that behavior.~~

The comparison is application-specific and should not be interpreted as evidence that machine-learning or nonlinear valuation models are intrinsically more regressive than linear models. Its purpose is to determine whether the predictive advantages of the more flexible baseline are accompanied by stronger departures from vertical neutrality under the common evaluation framework.

If the more flexible baseline provides meaningful predictive or ratio-dispersion gains while exhibiting a stronger value-related vertical-equity pattern, the model-design question is not simply whether to return to the linear benchmark. It is whether those predictive advantages can be retained while reducing the value-related pattern through the training objective.

~~Rather than equating any one of those metrics with the training objective, the method uses the identity $e_i(f) = \log r_i(f)$ together with $y_i = \log P_i$ to define a tractable empirical first-order target.~~

~~$$\widehat{\text{Cov}}(e(f), y) = \widehat{\text{Cov}}(\log r(f), \log P).$$~~

~~This covariance is a trainable first-order proxy for value-related residual structure, not an assessor-standard definition of vertical equity. A covariance near zero can coexist with nonlinear, distributional, or localized value-related patterns. Section 3 therefore uses covariance as the training mechanism while the broader assessor-facing and robustness suite above provides the independent validation of whether that mechanism actually reduces regressivity out of sample.~~

The assessor-facing diagnostics above define how the resulting valuations are evaluated; they need not be optimized directly during training. By Eqs. (1) and (4),

$$e_i(f) = \log r_i(f), \quad y_i = \log P_i.$$

A first-order value-related pattern can therefore be represented on the training scale through

$$\widehat{\text{Cov}}(e(f), y) = \widehat{\text{Cov}}(\log r(f), \log P). \quad (22)$$

Covariance is thus used as a tractable training proxy rather than as a replacement for the assessor-facing vertical-equity measures. Section 3 develops this correction.

3 Covariance-Guided Regressivity Correction

This section translates the value-related regressivity identified in Section 2 into a training-time correction penalty. We first define a global first-order covariance-based regressivity target on the log-price scale, then introduce a direct squared-covariance penalty, and finally derive a sample-additive upper-bound surrogate that is compatible with standard boosting software. The correction penalty leaves the prediction target, feature information, and base model class unchanged. That is, for a fixed penalty formulation, the only additional tuning parameter is the penalty strength, which we denote ρ .

Let $J_0(f)$ denote the baseline training objective, whose data-fit component is the squared log-price loss $L(f)$ defined in Section 2.2. All quantities below are computed from the “current” training sample. In temporal validation, the centering quantities and penalties are recomputed using the training block of each fold only.

3.1 Log-Scale Regressivity Target

Let

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i, \quad c_i = y_i - \bar{y}, \quad \bar{e}(f) = \frac{1}{n} \sum_{i=1}^n e_i(f). \quad (23)$$

After the final empirical run, replace this note with one concise results paragraph. Describe only the comparisons needed for the motivation: (i) the change in predictive performance, (ii) the change in aggregate ratio dispersion, (iii) assessment level as a separate dimension, and (iv) whether the complementary vertical-equity measures and the value-group profiles show a stronger regressive pattern under either baseline. Do not narrate every entry in Table 2.

Once the final results are populated, make the next statement declarative and consistent with the evidence in Table 2 and Figure 1.

The empirical covariance between log-residuals and log-prices is

$$C(f) = \widehat{\text{Cov}}(e(f), y) = \frac{1}{n} \sum_{i=1}^n (e_i(f) - \bar{e}(f))(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n e_i(f) c_i. \quad (24)$$

The last equality follows from $\sum_{i=1}^n c_i = 0$, so centering the residuals does not change the covariance.

This covariance has a direct slope interpretation. Let

$$V_y = \frac{1}{n} \sum_{i=1}^n c_i^2.$$

Provided $V_y > 0$, the least-squares slope from regressing log assessment ratios on log prices is

$$\text{slope}(e(f) \sim y) = \frac{C(f)}{V_y}. \quad (25)$$

Because V_y is fixed by the training sample, reducing $|C(f)|$ is equivalent to reducing the magnitude of this global linear trend. In particular, $C(f) < 0$ indicates that log assessment ratios tend to decline with log sale price, which is the first-order pattern associated with regressivity.

This slope is a training-scale mechanism measure, not the assessor-facing PRB statistic defined in Section 2.3. We use covariance rather than correlation because V_y is fixed within the training sample, whereas correlation additionally normalizes by the model-dependent residual variance. Covariance therefore provides a direct training target for the global value-related slope.

Sale price enters this construction as the observed market-value benchmark for verified training sales; it is not introduced as a protected attribute. The broader connection to dependence-based fair regression is discussed in Section 1.1.

3.2 Direct Squared-Covariance Penalty

We first penalize the chosen first-order target directly. The *covariance-penalized learner* is

$$f_\rho^{\text{cov}} \in \arg \min_{f \in \mathcal{H}} \left\{ J_0(f) + \frac{\rho}{2} C(f)^2 \right\}, \quad \rho \geq 0. \quad (26)$$

When $\rho = 0$, formulation (26) reduces to the baseline learner. As ρ increases, the optimization places greater weight on reducing the global residual-price trend. Squaring the covariance treats negative and positive trends symmetrically. This is important because the objective is not to replace regressivity with progressivity, but to move the first-order relationship toward zero.

For interpretation, it is enough to consider the contribution of the covariance penalty to the gradient of observation i :

$$\frac{\partial}{\partial \hat{y}_i} \left[\frac{\rho}{2} C(f)^2 \right] = \frac{\rho}{n} C(f) c_i. \quad (27)$$

Suppose, for example, that the fitted model is regressive due to $C(f) < 0$. For a sale below the geometric-mean price (lower-tier property), $c_i < 0$, and thus the covariance component of the gradient is positive, i.e., a descent step lowers its predicted value. For a sale above the geometric-mean price (higher-tier property), $c_i > 0$, and thus the covariance component is negative, i.e., a descent step raises its predicted value. The penalty therefore acts in the direction required to flatten the global log-ratio-log-price trend. Note that the direction and interpretation reverse automatically when the fitted trend is progressive.

Note that the direct penalty targets how assessment ratios vary with sale price, but not their overall assessment level. Multiplying every predicted value by the same positive constant shifts all log assessment ratios by the same amount but leaves the covariance $C(f)$ unchanged. Thus, even a model with $C(f) = 0$ can have a median or mean assessment ratio different from one. Hence, assessment level must be evaluated separately.

For second-order models (e.g., LightGBM), the corresponding Hessian of the squared log-price loss plus direct penalty is

$$\nabla_{\hat{y}}^2 \left(L(f) + \frac{\rho}{2} C(f)^2 \right) = \frac{2}{n} I + \frac{\rho}{n^2} c c^\top. \quad (28)$$

The rank-one Hessian term $\frac{\rho}{n^2}cc^\top$ introduces cross-observation curvature. Standard LightGBM custom objectives provide one gradient and one Hessian value per observation, so this dense term cannot be represented exactly through the usual interface. An exact second-order implementation therefore requires additional machinery: using only the diagonal is possible, but discards the cross-observation curvature. The next subsection proposes an alternative penalty term to handle this caveat. Further, the model-specific derivations are given in Appendix E.

3.3 Sample-Additive Covariance Upper-Bound Surrogate

For standard second-order boosting software, we seek a penalty that decomposes exactly across observations. Given that (24) already gives $C(f) = \frac{1}{n} \sum_{i=1}^n e_i(f)c_i$, we can make use of Jensen’s inequality that yields a sample-additive upper bound to the squared-covariance term directly.

$$C(f)^2 = \left(\frac{1}{n} \sum_{i=1}^n e_i(f)c_i \right)^2 \leq \frac{1}{n} \sum_{i=1}^n e_i(f)^2 c_i^2. \quad (29)$$

This inequality motivates the fully observation-additive surrogate penalty

$$\Psi^{\text{surr}}(f) = \frac{1}{n} \sum_{i=1}^n e_i(f)^2 c_i^2 = \frac{1}{n} \sum_{i=1}^n e_i(f)^2 (y_i - \bar{y})^2. \quad (30)$$

The resulting *surrogate-penalized learner* is therefore

$$f_\rho^{\text{surr}} \in \arg \min_{f \in \mathcal{H}} \{J_0(f) + \rho \Psi^{\text{surr}}(f)\}. \quad (31)$$

Thus, $\Psi^{\text{surr}}(f)$ is a genuine upper bound on the squared empirical covariance in (24). Nevertheless, the bound does not make surrogate minimization equivalent to direct covariance minimization: the upper bound can be loose when the signed contributions $e_i(f)c_i$ cancel in aggregate covariance. The surrogate instead penalizes the squared magnitude of each contribution before aggregation, so opposite-signed contributions cannot cancel within the penalty.

Combining the baseline loss with the surrogate gives

$$L(f) + \rho \Psi^{\text{surr}}(f) = \frac{1}{n} \sum_{i=1}^n [1 + \rho(y_i - \bar{y})^2] e_i(f)^2. \quad (32)$$

Thus, the surrogate is equivalent to a weighted squared-error objective with fixed observation weights

$$w_i(\rho) = 1 + \rho(y_i - \bar{y})^2. \quad (33)$$

Properties near the center of the log-price distribution receive weights close to the baseline weight, whereas properties in the lower and upper tails receive larger weights.

Since $\exp(\bar{y})$ is the geometric mean sale price, the weight function is symmetric in log-distance from this center: prices that differ from the geometric mean by reciprocal multiplicative factors receive the same additional weight. This symmetry concerns the weighting function only: the empirical lower and upper tails, and the resulting prediction changes, need not be symmetric.

This representation clarifies the mechanism without imposing a directional assumption on individual observations. The surrogate raises the cost of residuals farther from the center of the log-price distribution. Because it uses $e_i(f)^2 c_i^2$, however, it does not encode whether the aggregate residual–price trend is regressive or progressive. Its effect on covariance and on assessor-facing vertical-equity diagnostics must therefore be evaluated out of sample.

For squared loss, the observation-wise gradient and Hessian are

$$g_i = \frac{2}{n} e_i(f) [1 + \rho(y_i - \bar{y})^2], \quad (34)$$

$$h_i = \frac{2}{n} [1 + \rho(y_i - \bar{y})^2]. \quad (35)$$

Both quantities depend only on observation i once $\bar{y}$ has been computed. They can therefore be supplied through a standard second-order boosting custom-objective interface.

The implementation follows LightGBM’s observation-wise derivative convention. The exact numerical scaling of ρ is therefore tied to the implemented objective parameterization and is documented with the empirical specification.

Unlike the direct covariance penalty, the surrogate is not invariant to a common shift in all log predictions because it penalizes the magnitude of $e_i(f)$ itself. It can therefore affect overall assessment level in addition to the value-related trend.

3.4 Interpretation, Validation, and Scope

The two formulations serve complementary roles. The direct penalty targets the chosen global first-order association between log assessment ratios and log sale prices. The surrogate replaces direct covariance minimization with a sample-additive upper-bound objective that is directly compatible with standard boosting software. Because the two penalties have different scales, the numerical values of ρ are not directly comparable across formulations. Their appropriate scale can also vary across training samples and jurisdictions, so ρ is calibrated separately through chronological validation.

Neither formulation directly optimizes PRD, PRB, VEI, COD, or assessment level. Nor does zero training covariance imply the absence of nonlinear, geographic, property-class, temporal, or other local value-related patterns. The training covariance is therefore treated as a mechanism target rather than as a sufficient measure of vertical equity. Predictive performance, assessment level, PRD, PRB, VEI, ratio curves, and subgroup diagnostics remain separate validation and out-of-sample evaluation criteria.

Sale price is required to construct the correction only for verified sales used during model fitting. Once a model has been trained, the resulting valuation function continues to generate predictions from x alone, as in the baseline AVM; the correction does not require a current sale price for an unsold property.

Section 4 specifies the chronological validation and penalty-selection procedure used in the empirical application. Appendix D provides complementary intuition for the direct penalty: within a fixed linear prediction space, increasing ρ shrinks the targeted covariance smoothly while the increase in squared error is second order near $\rho = 0$. This result is an interpretive benchmark, not a guarantee for fully retrained boosted trees, whose splits, leaves, and stopping behavior can change with the penalty.

4 Application and Evaluation Design

4.1 Application to the Translated CCAO Residential Pipeline

For the empirical analysis, we implement in Python the components of CCAO’s residential valuation workflow required for model development and evaluation. The implementation closely follows CCAO’s data-preparation, model-fitting, and ratio-study procedures, preserving the operational structure of the existing workflow while enabling controlled evaluation of alternative training objectives.

The proposed methodology modifies only the model-training objective. We hold fixed the sale filters, predictor set, temporal data splits, transformation of predictions to the price scale, ratio-study procedures, and surrounding review process. This restricted intervention serves two purposes. First, it isolates the effect of incorporating vertical equity into model estimation. Second, it demonstrates how the methodology can be integrated into an existing mass-appraisal workflow without redesigning the broader valuation system. The CCAO application uses verified Cook County residential sales from the CCAO 2025 assessment-cycle data. Following the office pipeline, we exclude multicard PINs and sales marked as outliers. The pre-2024 modeling sample contains 349,661 observations. Sales from 2016–2022 form the development period; 2023 is held out for final testing. The 2024 sales block is reserved for later assessment-year analysis and is not used for the results reported here.

The model uses 95 predictors, including property characteristics, location and proximity variables, Census measures, time variables, parcel-shape measures, neighborhood identifiers, and other assessor variables. Twenty-three are treated as categorical. All predictors are constructed upstream; the penalized models use the same feature set as the LightGBM baseline.

Table 3: CCAO application and evaluation design.

Item	Design
Population	Verified Cook County residential sales
Development period	2016–2022
Untouched test period	2023
Pre-2024 observations	349,661
Predictors	95 total; 23 categorical
Validation	Eight expanding-window rolling-origin folds
Base nonlinear learner	LightGBM on log sale price
Primary correction	Covariance-guided sample-additive surrogate
Selection	Best validation accuracy among candidates meeting active safeguards

Within 2016–2022, we use eight expanding-window rolling-origin folds. Each validation block follows its training block in time. For the principal CCAO comparison, candidates are screened using the average validation PRD reference band and then ranked by predictive performance. PRB, VEI, assessment level, and dispersion are reported as additional diagnostics rather than used to select the displayed model. The 2023 test set is evaluated only after selection.

4.2 Required Comparators

Four comparisons are needed to interpret the method.

Linear and nonlinear baselines. Linear regression shows the performance of a simpler model; unpenalized LightGBM is the relevant modern AVM baseline.

Direct and surrogate penalties. The direct penalty tests the stated covariance target. The surrogate tests the implementation recommended for standard boosting software. A custom-objective fit at $\rho = 0$ is also needed because LightGBM initialization and training behavior can differ between an ordinary regression objective and a user-supplied objective. This control separates the effect of the penalty from the effect of entering the custom-objective code path.

Post-hoc rescaling. A one-parameter rescaling fitted on training or validation data is a low-cost competitor. It must be reported because a global correction may reproduce part of the penalty’s effect without changing the training objective. The paper therefore does not treat improvement over unpenalized LightGBM as sufficient evidence that custom regularization is necessary.

A constant multiplicative level adjustment and a slope-changing rescaling serve different roles. Multiplying every price prediction by a constant adds a constant to every log residual and therefore leaves $\text{Cov}(e, y)$ unchanged. By contrast, a one-degree calibration that changes the spread of predictions can alter the ratio–value slope and is the relevant competitor in the multi-county benchmark.

Richer information. Spatial, temporal, neighborhood, and property-characteristic improvements can reduce omitted structure and may improve both accuracy and equity. They are complements to the penalty, not substitutes assumed away by the paper. The preferred sequence is to build the strongest defensible AVM first and then test whether an explicit vertical-equity safeguard still adds value.

4.3 Exploratory Six-County ATTOM Benchmark

We also run a common ATTOM-derived pipeline in Cook County, IL; Allegheny County, PA; Maricopa County, AZ; King County, WA; Miami-Dade County, FL; and Middlesex County, MA. The benchmark contains 1,231,194 qualified single-family sales and 246,241 held-out sales. Each county uses chronological

validation and test blocks, the same modeling code path, and county-specific penalty values calibrated to the baseline prediction scale.

This is an exploratory transfer benchmark, not an audit of the six assessor offices. The data are commercially harmonized and do not reproduce each jurisdiction’s sales-verification rules, local features, legal setting, or production review. The benchmark was developed iteratively, and its present county configurations were not all frozen under one confirmatory protocol before test evaluation. We therefore use it to examine direction, heterogeneity, and failure modes. It does not provide confirmatory estimates of performance in the six jurisdictions.

4.4 Validation-Based Model Selection

The penalty parameter ρ governs an accuracy–equity tradeoff whose appropriate value cannot be determined from the training objective alone. A very small value may leave the baseline trend largely unchanged, whereas a very large value may distort predictions or overfit the equity pattern in the training period. We therefore treat ρ as a model-selection parameter and choose it using chronological validation.

Let $\mathcal{C}$ be a finite set of candidate configurations containing the unpenalized baseline and a prespecified grid of positive penalty values. The candidate set may include both the direct and surrogate formulations. Let $\mathcal{V}$ index the temporal validation folds, and let $\alpha_v \geq 0$ denote prespecified fold weights satisfying

$$\sum_{v \in \mathcal{V}} \alpha_v = 1.$$

For candidate $\eta \in \mathcal{C}$, let $\text{Perf}_v(\eta)$ be a predictive-performance score on fold v , oriented so that larger values are preferred. Let $Q_{m,v}(\eta)$ be the value of ratio-study statistic m on that fold, and let $[\underline{\kappa}_m, \bar{\kappa}_m]$ be its prespecified validation range. If $\mathcal{M}$ denotes the set of active safeguards, the default selection rule is

$$\begin{aligned} \eta^* &\in \arg \max_{\eta \in \mathcal{C}} \sum_{v \in \mathcal{V}} \alpha_v \text{Perf}_v(\eta) \\ \text{s.t. } &\underline{\kappa}_m \leq \sum_{v \in \mathcal{V}} \alpha_v Q_{m,v}(\eta) \leq \bar{\kappa}_m, \quad m \in \mathcal{M}. \end{aligned} \tag{36}$$

This formulation mirrors the operational decision problem: among models satisfying the active ratio-study safeguards, select the one with the strongest out-of-time predictive performance. It avoids collapsing prediction, assessment level, dispersion, and vertical equity into a single weighted score whose interpretation would depend on arbitrary scaling choices.

The averaged constraints in (36) allow a temporary deviation in one validation period to be offset by stronger performance in another. When stability across reassessment periods is a primary concern, a stricter rule can instead require

$$\underline{\kappa}_m \leq Q_{m,v}(\eta) \leq \bar{\kappa}_m, \quad m \in \mathcal{M}, \quad v \in \mathcal{V}.$$

The choice between average-fold and every-fold safeguards should be made before the final test period is examined. Likewise, the candidate grid, active statistics, reference ranges, fold weights, and any tie-breaking rule should be prespecified.

The selection procedure deliberately distinguishes between statistics used as constraints and statistics reported for evaluation. An office may, for example, use PRD as an active validation safeguard while reporting PRB, VEI, COD, assessment level, and subgroup ratio curves as additional diagnostics. The empirical design in the next section specifies the active safeguards used in the CCAO application.

If no candidate satisfies the prespecified constraints, formulation (36) returns no feasible model. This outcome should prompt examination of the data, base learner, candidate set, and source of the conflicting diagnostics rather than an unreported relaxation of the safeguards. Pareto-based rules and convex stacking can be useful for exploratory analysis of such cases, but they introduce additional choices and are therefore treated as secondary extensions in Appendix H. The final test period remains untouched until the complete validation-based selection procedure has been applied.

Either correctly connect this or move to appendix, and just reference it

Table 4: Principal CCAO 2023 test-set results. The penalized model was selected using rolling-origin validation, not the test set.

Model	ρ	R^2	MAE	Median r	Mean r	COD	COV	PRD	PRB	VEI
Linear regression	–	0.762	\$86,244	1.001	1.050	25.01%	36.86%	1.039	-0.011	-13.53%
LightGBM baseline	–	0.885	\$71,912	0.944	1.021	23.07%	35.48%	1.085	-0.118	-37.64%
Validation-selected surrogate	5.96	0.881	\$74,416	0.937	0.978	22.80%	33.64%	1.039	-0.014	-9.27%

5 Results

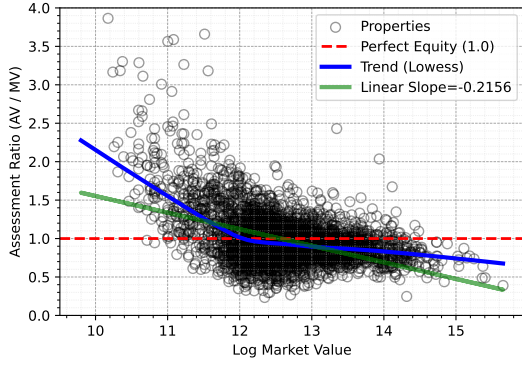
5.1 Accuracy Alone Does Not Remove the CCAO Ratio Trend

Table 4 reports the principal 2023 test-set comparison. LightGBM improves R^2 from 0.762 to 0.885 and lowers MAE from about \$86,244 to \$71,912 relative to linear regression. COD also falls. The vertical-equity measures move in the opposite direction: PRD rises to 1.085, PRB falls to -0.118 , and VEI falls to -37.64% .

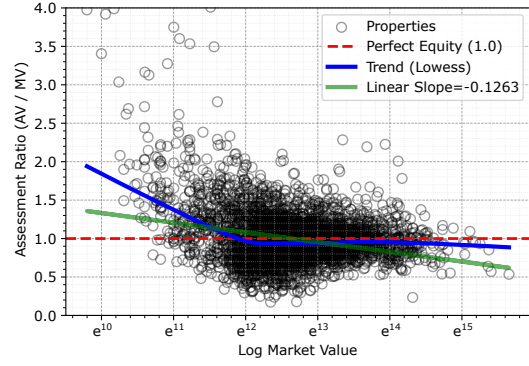
The validation-selected surrogate model uses $\rho = 5.96$. Relative to the LightGBM baseline, it reduces the PRD deviation from one from 0.085 to 0.039, the absolute PRB from 0.118 to 0.014, and the absolute VEI from 37.64% to 9.27%. R^2 decreases by 0.004, and MAE increases by about 3.5%. COD and COV improve modestly. The test-set PRD point estimate remains above the 1.03 reference bound. The distinction is substantive: the selected model improves the ratio pattern but does not satisfy every test-set reference band. The current CCAO results table does not include a custom-objective zero-penalty control. The comparison therefore measures the total change from the ordinary LightGBM fit to the implemented penalized fit; it does not isolate the penalty from every difference introduced by the custom-objective training path.

5.2 The Correction Acts in the Intended Direction

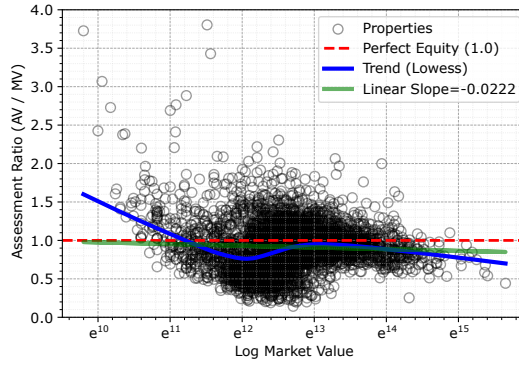
Figure 2 compares the baseline ratio curve with representative direct and surrogate corrections. The baseline curve declines sharply with value. Both penalized fits flatten the global trend, with the surrogate producing the largest reduction in this CCAO run. These plots are mechanism checks, not substitutes for PRD, PRB, VEI, assessment-level statistics, or subgroup analysis.



(a) Baseline LightGBM



(b) Direct squared-covariance penalty



(c) Covariance-guided surrogate

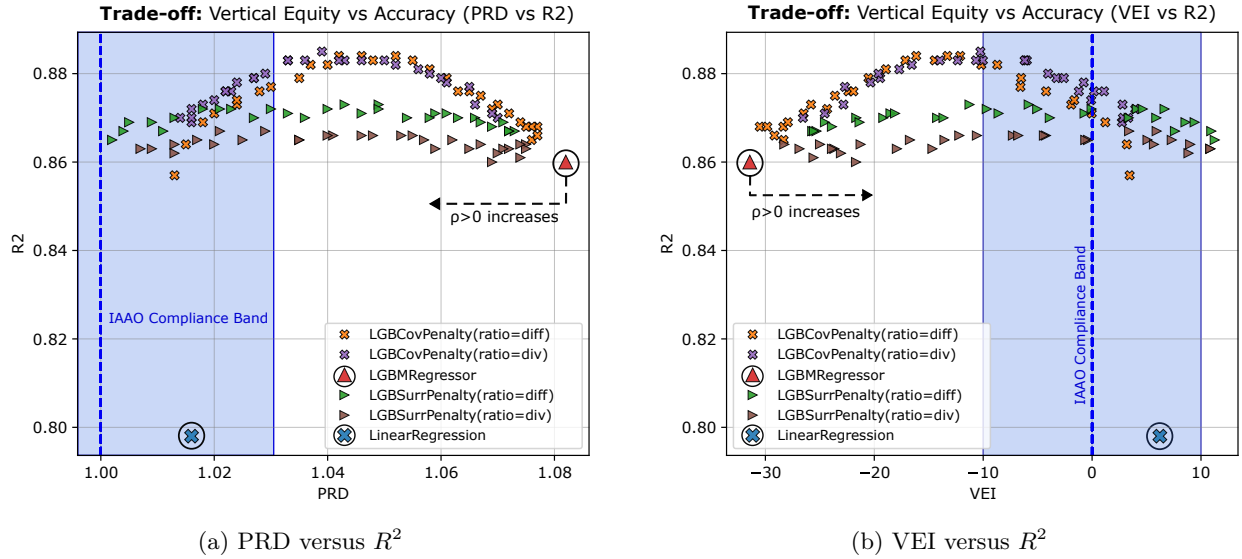
Figure 2: Assessment-ratio patterns in the translated CCAO test set. The penalty is intended to flatten the global value-related trend; all assessor-facing measures remain necessary for evaluation.

5.3 The Penalty Traces a Usable Validation Path

As ρ increases, the candidate models move from the accuracy-only LightGBM solution toward flatter ratio curves. The change is not a fixed or necessarily monotone tradeoff across all metrics. A moderate range gives large improvements in PRB and VEI with small changes in R^2 ; larger penalties eventually increase predictive error and can move the model toward progressivity. This pattern supports a grid search with temporal validation rather than a single theoretical value of ρ .

Table 5: PLACEHOLDER OF THE 3 REPRESENTATIVE PENALIZED RESULTS

Model Family	ρ	R^2	MAE	Med. r	Mean r	COD	COV	VEI	PRD	PRB
Linear Regression	–	0.762	\$86,256.879	1.001	1.051	25.013%	36.825%	-13.474%	1.039	-0.011
LGBMRegressor	–	0.883	\$72,063.853	0.944	1.021	23.106%	35.398%	-36.816%	1.085	-0.118
LGBSurrogatePenalty	0.100	0.881	\$73,271.882	0.936	1.011	23.125%	35.236%	-36.367%	1.084	-0.115
LGBSurrogatePenalty	0.189	0.881	\$73,208.883	0.936	1.010	23.042%	35.048%	-33.608%	1.080	-0.107
LGBSurrogatePenalty	0.404	0.882	\$73,110.847	0.937	1.006	22.924%	34.782%	-29.878%	1.075	-0.095
LGBSurrogatePenalty	0.761	0.883	\$72,974.386	0.937	1.001	22.749%	34.189%	-23.789%	1.067	-0.076
LGBSurrogatePenalty	1.628	0.882	\$73,233.014	0.938	0.994	22.645%	33.664%	-16.431%	1.057	-0.052
LGBSurrogatePenalty	3.070	0.881	\$73,663.987	0.937	0.986	22.667%	33.620%	-12.630%	1.048	-0.032
LGBSurrogatePenalty	6.572	0.882	\$74,428.114	0.937	0.976	22.806%	33.386%	-8.545%	1.038	-0.011
LGBSurrogatePenalty	12.390	0.879	\$75,846.355	0.935	0.968	23.249%	33.684%	-7.446%	1.032	0.003
LGBCovPenalty	0.100	0.879	\$73,459.227	0.937	1.014	23.210%	35.551%	-38.368%	1.087	-0.122
LGBCovPenalty	0.189	0.880	\$73,311.803	0.939	1.015	23.221%	35.443%	-37.671%	1.086	-0.119
LGBCovPenalty	0.404	0.882	\$72,947.453	0.941	1.017	23.224%	35.432%	-36.501%	1.083	-0.115
LGBCovPenalty	0.761	0.883	\$72,607.948	0.944	1.019	23.157%	35.246%	-34.017%	1.078	-0.106
LGBCovPenalty	1.628	0.886	\$72,198.279	0.949	1.023	23.176%	35.086%	-30.427%	1.071	-0.096
LGBCovPenalty	3.070	0.884	\$72,293.558	0.955	1.027	23.216%	35.053%	-26.850%	1.065	-0.085
LGBCovPenalty	6.572	0.881	\$72,871.805	0.961	1.030	23.282%	35.040%	-20.525%	1.055	-0.068
LGBCovPenalty	12.390	0.878	\$73,449.207	0.963	1.030	23.261%	34.757%	-16.050%	1.048	-0.057
LGBCovPenalty	26.519	0.870	\$74,741.535	0.963	1.028	23.310%	34.542%	-13.572%	1.041	-0.046
LGBCovPenalty	50.000	0.866	\$75,574.792	0.960	1.023	23.357%	34.522%	-11.105%	1.037	-0.039

Figure 3: CCAO accuracy–equity paths across penalty values. Reference bands guide candidate screening; the final test period is not used to choose ρ .

The exact regularization path derived in Appendix D explains why moderate penalties can remove a first-order component at second-order in-sample squared-error cost within a fixed prediction space. For fully retrained boosted trees, however, split structure and early stopping can change. The theory is used to interpret and scale the grid, [not to replace out-of-time validation](#).

5.4 Exploratory Six-County Evidence Shows Heterogeneity and Limits

Table 6 shows the unpenalized ATTOM benchmark results. Every county has at least one vertical-equity point estimate outside the reference bands used in the benchmark, but the patterns are not identical. Cook,

Table 6: Unpenalized LightGBM results in the six-county ATTOM benchmark. These are held-out model-benchmark point estimates, not official assessor results.

County	R^2 (log)	RMSE (log)	MAPE	COD	PRD	PRB	Median r
Cook, IL	0.833	0.283	0.207	21.199	1.083	-0.098	0.959
Allegheny, PA	0.779	0.325	0.253	26.129	1.103	-0.101	0.954
Maricopa, AZ	0.844	0.222	0.147	14.797	1.072	-0.055	0.989
King, WA	0.798	0.261	0.207	16.336	1.073	-0.028	1.106
Miami-Dade, FL	0.808	0.316	0.263	24.210	1.092	0.025	1.060
Middlesex, MA	0.780	0.264	0.191	18.565	1.085	-0.096	1.023

Allegheny, Maricopa, and Middlesex have negative PRB values outside the benchmark band. King and Miami-Dade have PRB values within the band even though their PRD values are high. Miami-Dade is especially useful as a boundary case: its positive PRB and curved ratio pattern show why a global covariance target and a single ratio statistic cannot be treated as complete descriptions of vertical equity.

Among the four counties whose baseline PRB point estimate is outside the benchmark band, at least one tested penalty places PRB inside the band in all four, with no more than a 1.84% increase in log-scale RMSE. In contrast, only Maricopa has a tested penalty that places both PRD and PRB point estimates inside their bands. These descriptive results suggest that the first-order correction can transfer across datasets, but they do not establish a universal or confirmatory compliance result.

5.5 The Current Evidence Does Not Establish Dominance over Recalibration

The multi-county benchmark includes validation-fitted centered recalibration maps of the baseline log predictions. These comparators can change the spread of predictions after model fitting and can therefore alter a global ratio-value trend. A constant multiplicative level adjustment is different: it shifts every log residual by the same amount and leaves the targeted covariance unchanged.

The existing comparisons do not support a general claim that the custom objective dominates the simpler recalibration. The in-processing penalty may still be useful when the target must influence tree construction or when the method will be extended to geographic, class-specific, or other structured constraints. For the current paper, the defensible rule is simpler: report both approaches, fit both without test data, and prefer recalibration when its out-of-time accuracy and equity results are materially the same.

5.6 Assessor Implementation

The method can be inserted into an existing AVM with the following sequence.

1. Build the strongest defensible baseline model using verified sales, appropriate time adjustment, relevant property features, and local market information.
2. Hold out a future period and create rolling or blocked validation folds that respect sale dates.
3. Report assessment level, COD, PRD, PRB, supplementary vertical-equity measures, and ratio curves for the baseline model.
4. Fit the covariance-guided objective over a logarithmic grid of ρ , including $\rho = 0$ as an implementation control.
5. Fit a simple post-hoc rescaling benchmark using development data only.
6. Select the most accurate candidate that meets prespecified validation safeguards, then evaluate it once on the untouched test period.

Table 7: Different pipeline components address different problems.

Component	Primary role
Better property, spatial, and temporal data	Reduce omitted structure and improve the conditional valuation model; may improve accuracy and equity together
Constant level calibration	Move the overall ratio level toward its required target; does not change log-residual covariance
One-parameter slope/spread rescaling	Low-cost post-hoc correction of a global value trend; required comparator for the penalty
Covariance-guided regularization	Discourage a global first-order log-ratio/log-price trend during model training
Stratified and local diagnostics	Detect offsetting geographic, class-specific, or nonlinear patterns hidden by aggregate measures
Temporal validation and uncertainty analysis	Test whether the selected correction generalizes across market periods

7. Repeat the diagnostics by geography, property class, value range, and time. Reject a global improvement that hides material local deterioration.

The office should save the penalty definition, grid, validation predictions, active constraints, selected configuration, and all ratio-study outputs. This documentation makes the change auditable and allows the correction to be re-evaluated in later assessment cycles.

6 Discussion and Conclusions

[The CCAO application answers a limited question.](#) A training-time correction can remove much of a countywide regressive ratio trend while retaining most of the predictive performance of the ordinary LightGBM model. The selected sample-additive model improves all three reported vertical-equity point estimates, although test PRD remains outside its reference band and MAE increases. [The results therefore support use of the penalty as a candidate model](#), not as proof that the equity problem has been solved.

The value of the approach is operational. It converts a ratio-study concern into a model-training option, works through a standard boosted-tree interface, and can be tuned with chronological validation. [Those features follow directly from the CCAO problem that motivated the collaboration.](#)

The evidence also limits the paper’s claim. Better data may improve accuracy and equity together. The covariance target is countywide and first-order; it does not set assessment level, guarantee any ratio-study range, or remove local inequity. The current CCAO comparison lacks a custom-objective zero-penalty control, and the exploratory county benchmark does not establish confirmatory external validity. Post-hoc recalibration may reproduce part of the gain. For these reasons, the penalty should enter the same review process as any other model change and should never be treated as an equity certificate.

6.1 Limitations

[First, the CCAO application uses sold properties.](#) Results on sales do not establish performance for the full unsold inventory, particularly when the sale sample is not representative. [Second, the primary application is a Python translation](#) of the CCAO workflow rather than an official production run. Third, the penalty targets log-scale covariance, while PRD, PRB, and VEI are price-scale statistics with different functional forms. Fourth, the main CCAO comparison does not yet include the custom-objective zero-penalty control needed to isolate implementation effects. Fifth, the main results are point estimates; uncertainty and variation across folds should accompany an operational decision. Sixth, a countywide correction can average away local regressivity and progressivity. Seventh, the ATTOM benchmark is exploratory, uses commercially harmonized data, and does not reproduce the six offices’ institutional data or a single confirmatory model-selection protocol. Finally, the current evidence does not show that in-processing regularization is better than post-hoc recalibration.

6.2 Future Work

The next evaluation step is a clean ablation that compares the baseline model, better spatial and temporal information, the covariance correction, and their combination under the same temporal splits. Comparable property and neighborhood features must use only information available before the target sale. Preliminary tests of second- and third-degree global recalibration have shown little consistent improvement over a first-degree map, so higher-order corrections should remain secondary unless they add stable out-of-time gains.

The main methodological extension is structured covariance control by geography, property class, and time, with safeguards for small or unstable groups. That extension should be judged against stratified post-hoc recalibration and against a stronger base AVM. Further work should also add block-bootstrap uncertainty, worst-period validation, the custom-objective zero-penalty control, and evaluation on the full assessment inventory. Desk review by assessors is needed to determine whether the changed predictions remain understandable and operationally defensible.

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A Metric Details and Guidance Status

The 2013 IAAO Standard on Ratio Studies is the established standard cited for PRD and COD guidance in this paper [21]. The May 2026 revision of the ratio-studies document is an exposure draft. Measures introduced or revised there, including VEI, are reported as supplementary diagnostics [26]. Reference bands are used for candidate screening; the paper does not infer legal compliance from a point estimate.

The weighted mean ratio satisfies

$$\bar{r}_W = \frac{\sum_i r_i P_i}{\sum_i P_i} = \frac{\sum_i \hat{P}_i}{\sum_i P_i},$$

and PRD is $\bar{r}/\bar{r}_W$. PRB follows the IAAO median-normalized regression definition. The full implementation [used in the application should be released](#) with the code so that all transformations, trimming rules, and proxy definitions are reproducible.

B Squared-Error Prediction and Residual–Value Dependence

Let $Y = \log P$, let X be the observed feature vector, and let

$$f^*(X) = \mathbb{E}[Y \mid X]$$

be the squared-error Bayes predictor. Define $e^* = f^*(X) - Y$. If $\mathbb{E}[Y^2] < \infty$, then

Proposition 1.

$$\text{Cov}(e^*, Y) = -\mathbb{E}[\text{Var}(Y \mid X)] \leq 0.$$

Proof. Using the law of total variance,

$$\text{Cov}(e^*, Y) = \text{Cov}(\mathbb{E}[Y \mid X] - Y, Y) = \text{Var}(\mathbb{E}[Y \mid X]) - \text{Var}(Y) = -\mathbb{E}[\text{Var}(Y \mid X)].$$

□

This identity explains a source of negative log-residual dependence when important price variation remains unresolved after conditioning on X . It does not imply that every finite-sample model is regressive under every assessor metric. It also does not imply that adding features cannot improve both accuracy and equity.

C Covariance and PRD on the Price Scale

Let $P > 0$ and $r = \hat{P}/P > 0$. The population PRD can be written as

$$\text{PRD} = \frac{\mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]}.$$

Then

Proposition 2.

$$\text{sign}(\text{Cov}(r, P)) = \text{sign}(1 - \text{PRD}).$$

Proof. Since $\mathbb{E}[rP] > 0$,

$$1 - \text{PRD} = \frac{\mathbb{E}[rP] - \mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]} = \frac{\text{Cov}(r, P)}{\mathbb{E}[rP]}.$$

□

The penalty in the paper acts on $\text{Cov}(\log r, \log P)$, not on $\text{Cov}(r, P)$. Proposition 2 therefore gives directional motivation rather than an equality between the training target and PRD.

D Exact Fixed-Space Regularization Path

Let $\mathcal{V} \subseteq \mathbb{R}^n$ be a closed linear prediction space, define

$$\langle u, v \rangle_n = \frac{1}{n} u^\top v, \quad \|u\|_n^2 = \langle u, u \rangle_n, \quad c = y - \bar{y}\mathbf{1},$$

and consider

$$f_\rho \in \arg \min_{f \in \mathcal{V}} \left\{ \|f - y\|_n^2 + \frac{\rho}{2} \langle f - y, c \rangle_n^2 \right\}. \quad (37)$$

Let $P_\mathcal{V}$ be the orthogonal projection under $\langle \cdot, \cdot \rangle_n$ and set

$$f_0 = P_\mathcal{V}y, \quad C_0 = \langle f_0 - y, c \rangle_n, \quad g = P_\mathcal{V}c, \quad A = \|g\|_n^2.$$

Proposition 3 (Covariance shrinkage path). *If $A > 0$, then*

$$f_\rho = f_0 - \frac{\rho}{2} C_\rho g, \quad C_\rho = \frac{C_0}{1 + \rho A/2}.$$

Consequently, the slope of residuals on y is multiplied by

$$q(\rho) = \frac{1}{1 + \rho A/2}.$$

Proof. For every $h \in \mathcal{V}$, first-order optimality gives

$$2\langle f_\rho - y, h \rangle_n + \rho C_\rho \langle c, h \rangle_n = 0.$$

Since $f_0 = P_\mathcal{V}y$, $\langle f_0 - y, h \rangle_n = 0$. Writing $d_\rho = f_\rho - f_0 \in \mathcal{V}$ and $g = P_\mathcal{V}c$ gives

$$2d_\rho + \rho C_\rho g = 0.$$

Taking the inner product with c yields

$$C_\rho = C_0 - \frac{\rho}{2} A C_\rho,$$

which proves the result. □

Proposition 4 (Accuracy cost). *Under Proposition 3,*

$$\|f_\rho - y\|_n^2 - \|f_0 - y\|_n^2 = \frac{C_0^2}{A} (1 - q(\rho))^2.$$

Within a fixed prediction space, the targeted covariance falls to first order near $\rho = 0$, while the in-sample squared-error cost is second order. For boosted trees this is a local benchmark only. Retraining can change splits, leaves, and stopping iteration.

E Linear and Boosting Implementations

E.1 Model-Specific Implementations

This is from the
main body

We now instantiate the two penalties for the model classes studied in Section 2.2. Linear regression represents CCAO's earlier modeling framework and provides a transparent analytical benchmark. Second-order tree boosting represents the current LightGBM framework and the principal nonlinear implementation. These derivations also clarify the relevant computational distinction: the surrogate penalty is directly compatible with observation-wise custom-objective interfaces, whereas the direct penalty requires either a diagonal curvature approximation or a custom leaf-wise solver that retains its rank-one cross-observation curvature.

E.1.1 Linear Covariance-Penalized Models

Consider the linear hypothesis class

$$\mathcal{H}_{\text{lin}} = \{f_\theta : x \mapsto x^\top \theta : \theta \in \Theta \subseteq \mathbb{R}^p\},$$

where the feature vector may include an intercept. Let $X \in \mathbb{R}^{n \times p}$ be the corresponding design matrix and write

$$e = X\theta - y, \quad c = Hy = y - \bar{y}\mathbf{1}, \quad H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \quad D = \text{diag}(c_1^2, \dots, c_n^2).$$

Then $C(f_\theta) = n^{-1}e^\top c$, and the baseline loss in (5) is $L(f_\theta) = n^{-1}\|X\theta - y\|_2^2$.

Direct squared-covariance penalty. For the direct formulation, the linear estimator solves

$$\theta_\rho^{\text{cov}} \in \arg \min_{\theta \in \Theta} \left\{ \frac{1}{n} \|X\theta - y\|_2^2 + \frac{\rho}{2n^2} ((X\theta - y)^\top c)^2 \right\}. \quad (38)$$

Define

$$a = X^\top c, \quad b = y^\top c = c^\top c.$$

When $\Theta = \mathbb{R}^p$, the first-order optimality condition is

$$\left(X^\top X + \frac{\rho}{2n} aa^\top \right) \theta = X^\top y + \frac{\rho}{2n} ba. \quad (39)$$

Consequently, whenever the displayed inverse exists,

$$\theta_\rho^{\text{cov}} = \left(X^\top X + \frac{\rho}{2n} aa^\top \right)^{-1} \left(X^\top y + \frac{\rho}{2n} ba \right). \quad (40)$$

Thus, the direct penalty produces a rank-one modification of the ordinary least-squares normal equations.

Sample-additive surrogate. For the surrogate penalty, the linear estimator solves

$$\theta_\rho^{\text{surr}} \in \arg \min_{\theta \in \Theta} \left\{ \frac{1}{n} \|X\theta - y\|_2^2 + \frac{\rho}{n} (X\theta - y)^\top D (X\theta - y) \right\}. \quad (41)$$

When $\Theta = \mathbb{R}^p$, its first-order optimality condition is

$$(X^\top X + \rho X^\top D X) \theta = X^\top y + \rho X^\top D y, \quad (42)$$

and hence, whenever the displayed inverse exists,

$$\theta_\rho^{\text{surr}} = (X^\top X + \rho X^\top D X)^{-1} (X^\top y + \rho X^\top D y). \quad (43)$$

Equivalently, the surrogate is weighted least squares with diagonal weight matrix $I + \rho D$.

E.1.2 Covariance-Penalized Boosted Tree

We next specialize the penalties to the second-order boosting step in (??). At iteration m , let

$$f_i = f^{(m-1)}(x_i), \quad e_i = f_i - y_i, \quad z = C(f^{(m-1)}) = \frac{1}{n} \sum_{i=1}^n e_i c_i.$$

As in the standard boosting construction, the tree and its leaf weights are fitted to the current derivatives, after which the learning rate is applied in the ensemble update.

Direct penalty with diagonal curvature. Under the direct squared-covariance penalty, the gradient supplied to the boosting step is

$$g_i^{\text{cov}} = \frac{2}{n} e_i + \frac{\rho}{n} z c_i. \quad (44)$$

The exact Hessian is the dense matrix in (28), therefore it is not directly applicable in standard boosted tree packages.

One heuristic is to retain only its diagonal, which gives

$$h_i^{\text{cov,diag}} = \frac{2}{n} + \frac{\rho}{n^2} c_i^2. \quad (45)$$

Equations (44)–(45) define an observation-wise custom objective that can be supplied to standard second-order boosting software. The approximation preserves the exact gradient but ignores the cross-observation curvature generated by the direct penalty.

Sample-additive surrogate. For the surrogate objective, the observation-wise derivatives are

$$g_i^{\text{surr}} = \frac{2}{n} e_i (1 + \rho c_i^2), \quad (46)$$

$$h_i^{\text{surr}} = \frac{2}{n} (1 + \rho c_i^2). \quad (47)$$

Thus, once c_i^2 has been precomputed, the surrogate can be implemented directly through the standard gradient and Hessian interface used by LightGBM. This observation-wise implementation is the formulation used for the principal practical specification in the CCAO application.

E.2 Linear Model

Let $X \in \mathbb{R}^{n \times p}$ and $e = X\theta - y$. With

$$c = y - \bar{y}\mathbf{1}, \quad a = X^\top c, \quad b = y^\top c,$$

the direct penalized estimator under loss $\|X\theta - y\|_2^2/n$ satisfies

$$\left(\frac{1}{n} X^\top X + \frac{\rho}{2n^2} aa^\top \right) \theta = \frac{1}{n} X^\top y + \frac{\rho}{2n^2} ba.$$

This is a rank-one modification of the normal equations.

For the surrogate, let $D = \text{diag}(c_1^2, \dots, c_n^2)$. Then

$$\left(\frac{1}{n} X^\top X + \frac{\rho}{n} X^\top DX \right) \theta = \frac{1}{n} X^\top y + \frac{\rho}{n} X^\top Dy.$$

E.3 Direct Penalty in Boosting

For current predictions f , define $e = f - y$ and

$$z = \frac{1}{n} e^\top c.$$

Under squared loss, the direct objective has

$$g_i = \frac{2e_i}{n} + \frac{\rho}{n} z c_i,$$

and exact Hessian

$$\nabla_f^2 = \frac{2}{n} I + \frac{\rho}{n^2} c c^\top.$$

The diagonal approximation uses

$$h_i^{\text{diag}} = \frac{2}{n} + \frac{\rho}{n^2} c_i^2.$$

For a fixed tree with leaves $I_1, \dots, I_J$, let

$$b_j = \sum_{i \in I_j} e_i, \quad s_j = \sum_{i \in I_j} c_i, \quad n_j = |I_j|.$$

With leaf-weight vector w , ridge parameter λ , and $s = (s_1, \dots, s_J)^\top$, the exact local quadratic system is

$$A_T w^* = -g_T, \quad g_{T,j} = \frac{2b_j}{n} + \frac{\rho z s_j}{n},$$

$$A_T = \text{diag}\left(\frac{2n_1}{n} + \lambda, \dots, \frac{2n_J}{n} + \lambda\right) + \frac{\rho}{n^2} s s^\top.$$

Thus $w^* = -A_T^{-1} g_T$ when A_T is nonsingular. This solver retains the cross-observation curvature but requires a custom tree-construction procedure; it is not available through the usual observation-wise API.

E.4 Surrogate in Boosting

For

$$\Psi^{\text{sur}}(f) = \frac{1}{n} \sum_i e_i^2 c_i^2,$$

the derivatives are observation-wise:

$$g_i = \frac{2e_i}{n} + \frac{2\rho}{n} e_i c_i^2, \quad h_i = \frac{2}{n} + \frac{2\rho}{n} c_i^2.$$

This is the implementation used for the primary practical model.

F CCAO Reproducibility Requirements

A replication package for the CCAO application should contain the penalty grid, fold-level validation predictions, active selection safeguards, selected configuration, final test predictions, and assessor metrics before and after any separate level calibration. It should include both the ordinary LightGBM baseline and a custom-objective fit with zero penalty. Before operational use, the same analysis should report block-bootstrap intervals and ratio curves by township, property class, value range, and time. These outputs separate model-selection uncertainty, implementation effects, and local variation.

G Exploratory ATTOM Benchmark and Recalibration

The exploratory ATTOM benchmark uses single-family recorder transfers [that pass the benchmark price and data-quality filters](#). Each transfer is joined to the most recent assessor-history record available before the sale and to lagged contextual variables. The final 20% of eligible sales form a chronological test block, with a validation block inside the remaining data.

The benchmark compares the ordinary LightGBM fit, the custom-objective zero-penalty control, positive covariance-penalty values, and centered recalibration maps of the baseline log predictions. Recalibration parameters are estimated without using the test block. Constant level calibration is reported separately because it changes the overall ratio level but not the targeted log-residual covariance.

The current county analysis was developed iteratively and is treated as exploratory. A confirmatory version must freeze the feature set, LightGBM settings, penalty grid, recalibration family, and selection rule before opening any county’s test block. It must report the full non-dominated frontier for prediction error, PRD, PRB, COD, assessment level, and value-decile ratio curves. Until that protocol is rerun, the county results are evidence about possible behavior and failure modes rather than definitive external validation.

H Alternative Selection and Stacking Extensions

When no candidate satisfies every active validation safeguard, the office should first inspect the source of infeasibility rather than silently replacing the constraints with a weighted score. Possible actions are to improve the data, revise the candidate set, separate level calibration from vertical-trend control, or report the least-violating candidate without calling it feasible.

A normalized distance-to-ideal rule can be used for exploratory comparison. Convex stacking can also enlarge the attainable set when constraints such as PRD can be written as affine inequalities in the stacking weights. These methods are secondary because they add choices and can make the final model harder to explain. The discrete constrained rule in (36) remains the recommended default.

Data, Code, and Institutional Statement

[The primary application uses CCAO data](#) and a translated version of the office’s published residential modeling workflow. The multi-county benchmark uses licensed ATTOM-derived records that cannot be redistributed through the paper. Reproducibility therefore depends on archiving the custom objectives, preprocessing specifications, temporal splits, and evaluation code, together with instructions for authorized data users. The results reported here are research findings, not official CCAO assessment decisions. The paper makes no claim that CCAO has adopted the correction or placed it into production.